

P1.3 Energy and work

Learning objectives

After this topic, you should know:

- what work means in science
- how work and energy are related
- how to calculate the work done by a force
- what happens to work done to overcome friction.



Figure 1 Working out

Worked example

A builder pushed a wheelbarrow a distance of 5.0 m across flat ground with a force of 50 N. How much work was done by the builder?

Solution

work done = force applied $\times$ distance moved

$$= 50 \text{ N} \times 5.0 \text{ m}$$

$$= 250 \text{ J}$$



Figure 2 Pulling a lorry

Working out

In a fitness centre or a gym, you have to work hard to keep fit. Lifting weights and pedalling on an exercise bike are just two ways to keep fit. Whichever way you choose to keep fit, you have to apply a force to move something. So the work you do causes a transfer of energy.

When an object is moved by a force, **work** is done on the object by the force. So the force transfers energy to the object. The amount of energy transferred to the object is equal to the work done on it. For example, to raise an object, you need to apply a force to it to overcome the force of gravity on it. If the work you do on the object is 20 J, the energy transferred to it must be 20 J. So its gravitational potential energy store increases by 20 J.

energy transferred = work done

The work done by a force depends on the size of the force and the distance moved. One joule of work is done when a force of one newton causes an object to move a distance of one metre in the direction of the force. To calculate the work done by a force when it causes displacement of an object, use this equation:

$$\begin{array}{lll} \text{work done, } W & = & \text{force applied, } F \times \text{distance moved along the line} \\ \text{(joules, J)} & & \text{(newtons, N)} \quad \text{of action of the force, } s \\ & & \text{(metres, m)} \end{array}$$

Superhuman force

Imagine pulling a lorry over 40 m. On level ground, a pull force of about 2000 N is needed. The work done by the pulling force is 80 kJ (= 2000 N $\times$ 40 m). Very few people can manage to pull with such force. Don't even try it, though. The people who can do it are very, very strong and have trained specially for it.

Doing work

Carry out a series of experiments to calculate the work done in performing the tasks below. Use a newton-meter to measure the force applied, and use a metre ruler to measure the distance moved.

- 1 Drag a small box a measured distance across a rough surface.
- 2 Repeat the test above with two rubber bands wrapped around the box (Figure 3).

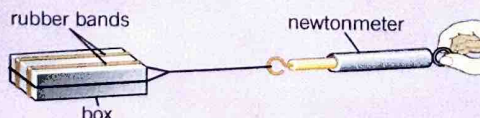


Figure 3 At work

- What is the resolution of your measuring instruments? Repeat your tests and comment on the precision of your repeat measurements. Can you be confident about the accuracy of your results?



Friction at work

Work done to overcome friction is mainly transferred to thermal energy stores by heating.

- 1 If you rub your hands together vigorously, they become warm. Your muscles do work to overcome the friction between your hands. The work you do is transferred as energy that warms your hands.
- 2 Brake pads on a vehicle become hot if the brakes are applied for too long. Friction between the brake pads and the wheel discs opposes the motion of the wheel. The force of friction does work on the brake pads and the wheel discs. As a result, energy is transferred from the kinetic energy store of the vehicle to the thermal energy store of the brake pads and the wheel discs. This makes them become hot and transfer energy by heating to the thermal energy store of the surrounding air.
- 3 Meteorites are small objects from space that enter the Earth's atmosphere and fall to the ground. As they pass through the atmosphere, friction caused by air resistance acts upon them. This results in energy being transferred from the meteorite's gravitational potential energy and kinetic energy stores to the meteorite's thermal energy store, causing the meteorite to heat up. If a meteorite becomes hot enough, it glows and becomes visible as a 'shooting star'. Very small objects can burn up completely. The surface of a space vehicle is designed to withstand the very high temperatures caused by this friction when it re-enters the Earth's atmosphere.

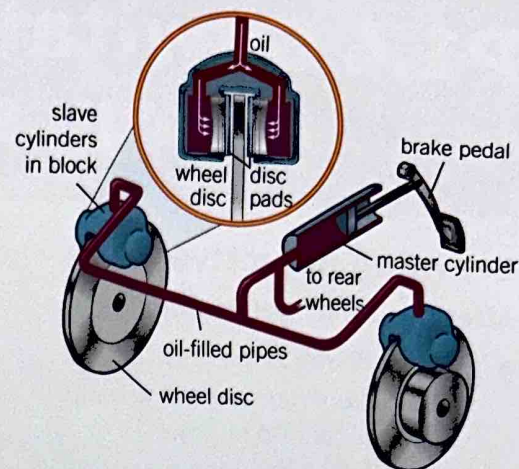


Figure 4 Disc brakes

- 1 a State what happens to the energy transferred:
 - i by a rower rowing a boat [1 mark]
 - ii by an electric motor used to raise a car park barrier. [1 mark]
 b Calculate how much work is done when a force of 3000 N pulls a truck through a distance of 50 m in the direction of the force. [1 mark]
- 2 A car is brought to a standstill when the driver applies the brakes.
 - a Explain why the brake pads become warm. [2 marks]
 - b The car travelled a distance of 20 metres after the brakes were applied. The braking force on the car during this time was 7000 N. Calculate the work done by the braking force. [1 mark]
- 3 a Calculate the work done when:
 - i a force of 20 N makes an object move 4.8 m in the direction of the force [1 mark]
 - ii an object of weight 80 N is raised through a height of 1.2 m. [1 mark]
 b When a cyclist brakes, his kinetic energy store is reduced from 1400 J to zero in a distance of 7.0 m. Calculate the braking force. [2 marks]
- 4 A student pushes a box at a steady speed a distance of 12 m across a level floor.
 - a The student applied a horizontal force of 25 N to the box. Calculate the work done by the student. [1 mark]
 - b Describe the energy transfers and changes to energy stores as the box moves. [3 marks]

Study tip

If you calculate the work done, this is equal to the energy transferred.

Key points

- Work is done on an object when a force makes the object move.
- Energy transferred = work done.
- Work done is $W = F s$ where F is the force and s is the distance moved (along the line of action of the force).
- Work done to overcome friction is transferred as energy to the thermal energy stores of the objects that rub together and to the surroundings.